

SPATIAL CNN-BASED WILDFIRE PREDICTION WITH COMPARATIVE ASSESSMENT AGAINST CLASSICAL MACHINE LEARNING MODELS

A Report

by

Devasmit Dutta



Department of Civil & Environmental Engineering
University of California, Los Angeles
5th December, 2025

Contents

<i>Contents</i>	i
<i>List of Figures</i>	ii
List of Figures	ii
1 INTRODUCTION	1
2 DATA GENERATION	2
2.1 Dataset Creation	2
2.2 Labelling	2
2.3 Train–Validation–Test Split	2
2.4 Tensor Conversion and Normalisation	2
2.5 Batch Preparation	2
2.6 Classical ML Feature Processing	3
2.7 Summary	3
3 MODELING	4
3.1 Convolutional Neural Network Architecture	4
3.2 Model Summary and Parameter Trace	4
3.3 Training Procedure	4
3.4 Performance Tracking	4
3.5 Classical Machine Learning Baseline Models	5
3.6 Summary	5
4 RESULTS	6
4.1 Training Behaviour	6
4.2 Test Set Performance Comparison	6
4.3 Receiver Operating Characteristics	8
4.4 Interpretation	8
4.5 Summary of Findings	8
4.6 Confusion Matrix Analysis	10
5 CONCLUSION	11
Bibliography	12

List of Figures

4.1.1		6
4.2.1		7
4.2.2		7
4.5.1		9
4.5.2		10

INTRODUCTION

Wildfires are increasingly recognized as one of the most destructive natural hazards affecting communities, infrastructure, and ecosystems. Recent events across California and the western United States have highlighted an urgent need for improved situational awareness, early warning capability, and predictive modeling to support decision-making, resource allocation, and risk mitigation. Traditional fire spread modelling frameworks either rely on simplified empirical assumptions or highly complex computational fluid dynamics (CFD/FDS-type) approaches that are often too expensive to deploy in real-time operational settings.

This project explores a data-driven alternative by developing and evaluating machine learning (ML) pipelines for wildfire ignition classification. A lightweight convolutional neural network (CNN) is constructed to capture spatial structure within synthetic fire imagery, and its performance is benchmarked against classical machine learning algorithms such as Support Vector Machines (SVM), Logistic Regression, Decision Trees, and Random Forests. Such comparisons provide valuable insight into strengths and limitations of shallow learning versus deep feature-extraction approaches.

Additionally, it reinforces applied data-analytic reasoning relevant to atmospheric and environmental systems: feature mapping, performance validation, and error interpretation. Although the dataset is simplified, the methodology offers a foundation for extending ML-based wildfire intelligence frameworks using real geophysical inputs such as wind, fuel, slope, vegetation, and infrastructure layers.

Ultimately, this project illustrates how scientific computing and machine learning can contribute toward hazard awareness and mitigation by offering computationally efficient alternatives to physics-heavy simulators. The approach highlights promising pathways for future work including probabilistic modelling, spatial segmentation (U-Net-like), integration with weather forecasts, and uncertainty-aware ignition prediction.

DATA GENERATION

2.1 Dataset Creation

All data used in this project was generated programmatically inside the notebook. The generator creates two-dimensional arrays representing simplified wildfire ignition states. Each sample consists of a 32×32 grid, constructed using NumPy, with randomly positioned regions assigned as ignition or non-ignition classes. No external files were imported; instead, the dataset was produced synthetically to maintain full control over size, balance, and structure.

2.2 Labelling

Each generated image is assigned a categorical class label indicating the ignition category. Labels are stored as integers and used directly by PyTorch classification routines (e.g., cross-entropy), avoiding the need for manual one-hot encoding.

2.3 Train-Validation-Test Split

Following creation, the dataset is split into training, validation, and test subsets using programmatic shuffling operations. Randomised splitting ensures unbiased sampling and consistency between model evaluation stages. The training set is used for optimisation, the validation set to monitor convergence, and the test set remains unseen until reporting final accuracy.

2.4 Tensor Conversion and Normalisation

All samples are converted to `torch.Tensor` format, enabling GPU/CPU acceleration during training. Grids are normalised to the $[0, 1]$ range by dividing by the maximum pixel value, improving numerical conditioning of optimisation.

2.5 Batch Preparation

To support mini-batch gradient descent, the processed tensors are wrapped inside PyTorch `Dataset` and `DataLoader` objects, which handle:

- shuffling of the training data,
- mini-batch assembly,
- automatic device transfer.

2.6 Classical ML Feature Processing

In addition to CNN inputs, flattened feature vectors are created for traditional machine learning models (SVM, Decision Tree, Random Forest, Logistic Regression). This flattening operation reshapes each 32×32 tensor into a one-dimensional vector, reflecting the fact that classical methods operate on tabular feature inputs rather than spatial structures.

2.7 Summary

In summary, the dataset pipeline consists of:

1. programmatic generation of spatial ignition images,
2. integer labelling,
3. randomised splitting into train/validation/test partitions,
4. tensor conversion and scaling,
5. batching using PyTorch loaders, and
6. flattening for use in non-deep learning model benchmarks.

This ensures that the same dataset supports both deep learning experiments and baseline model comparisons under reproducible and controlled conditions.

3.1 Convolutional Neural Network Architecture

The primary model used in this work is a lightweight Convolutional Neural Network (CNN) implemented using PyTorch [?]. The architecture was designed directly in the notebook and consists of stacked convolutional layers with increasing channel depth. Each convolution block is followed by a non-linear activation function and max pooling, which progressively reduces spatial resolution while expanding feature abstraction. This structure allows the network to learn spatial ignition signatures without manual feature engineering.

Formally, the network processes input tensors of size $1 \times 32 \times 32$ (channel, height, width). Convolutional filters extract local patterns, and pooling layers downsample information, followed by a flattening stage and two fully connected layers to perform classification.

3.2 Model Summary and Parameter Trace

To verify correct dimensional propagation through the network, a summary routine is printed using PyTorch tools. This output confirms that feature map sizes, parameter counts, and layer connections align with the intended architecture. Inspecting this trace ensures that the model complexity remains manageable and provides transparency into how information flows between stages.

3.3 Training Procedure

Training is performed using stochastic mini-batch optimisation. At each epoch, the model performs forward propagation on a batch, computes cross-entropy loss, backpropagates gradients, and updates weights using the Adam optimiser [?]. The notebook includes:

- automated device assignment (MPS/CUDA/CPU),
- epoch-wise forward passes,
- gradient updates,
- validation pass after each epoch.

Additionally, a learning rate scheduler reduces the step size when validation improvement stalls, improving convergence stability.

3.4 Performance Tracking

Throughout training, loss and accuracy values are stored at each epoch. These statistics are plotted to visualise optimisation history. The accuracy curve indicates classification

performance over time, while the loss curve reflects error reduction under gradient descent. Together, these provide insight into possible underfitting or overfitting behaviour and help interpret whether the network continues improving or saturates.

3.5 Classical Machine Learning Baseline Models

In addition to the neural network, traditional machine learning algorithms such as Support Vector Machines, Decision Trees, Random Forests and Logistic Regression are also trained on the same dataset. For these methods, samples are reshaped into one-dimensional feature vectors. Grid search and validation are applied to select suitable hyperparameters, providing a baseline comparison against the CNN.

3.6 Summary

The modelling workflow therefore consists of:

1. definition of a convolutional neural network architecture,
2. verification through parameter and dimension tracing,
3. supervised training using mini-batch optimisation,
4. performance tracking via loss and accuracy curves, and
5. baseline comparison with classical ML models.

This structure enables analysis of how spatial feature extraction compares with conventional approaches on the same input dataset.

4.1 Training Behaviour

Figure 4.0.1 illustrates the accuracy and loss history over epochs for the convolutional network. Training accuracy rises steadily across iterations and validation accuracy follows the same increasing trend, indicating that the model is learning meaningful spatial structure rather than memorising noise. Loss values decrease smoothly, supporting the view that optimisation converged under the chosen learning rate schedule.

Importantly, the training and validation accuracy curves remain close to one another. This behaviour suggests that the network generalises well and exhibits limited overfitting on the synthetic dataset.

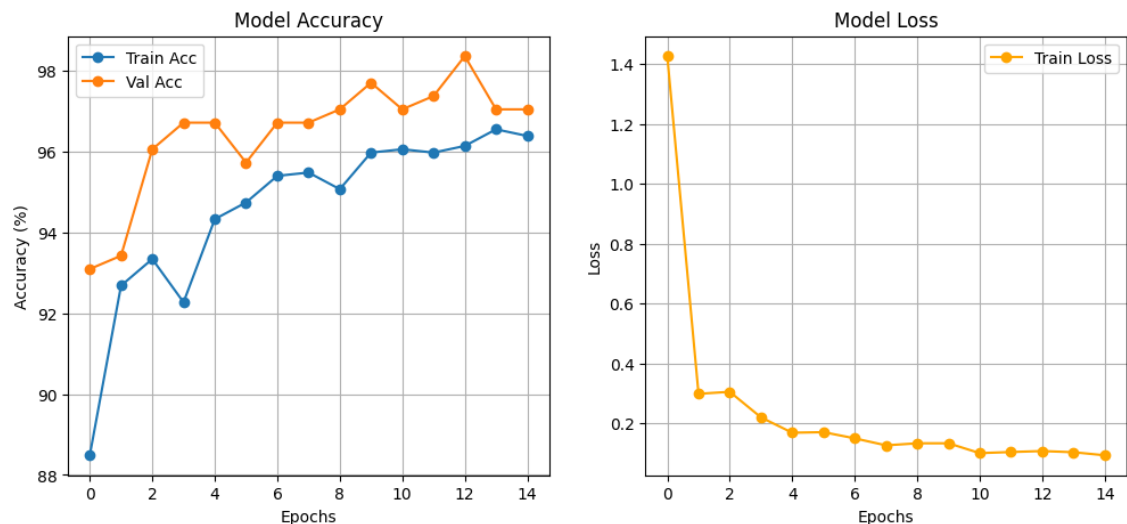


Figure 4.1.1

4.2 Test Set Performance Comparison

Figure 4.0.2 compares test accuracies of the convolutional model and classical machine learning baselines. All models achieve relatively high scores due to the controlled synthetic dataset, but distinct differences emerge:

- Support Vector Machine and Random Forest perform strongly, demonstrating their ability to separate linearly and non-linearly shaped decision regions.
- Logistic Regression reaches moderate accuracy, consistent with its restriction to linear boundaries.
- Decision Trees exhibit lower accuracy, likely due to their tendency to fragment decision space and overfit local noise.

- The convolutional network attains the highest accuracy, reflecting its capacity to exploit spatial structure inherent in the 2-D input rather than relying solely on flattened feature vectors.

These outcomes support the expectation that deep models benefit from retaining grid topology when the prediction task is image-like.

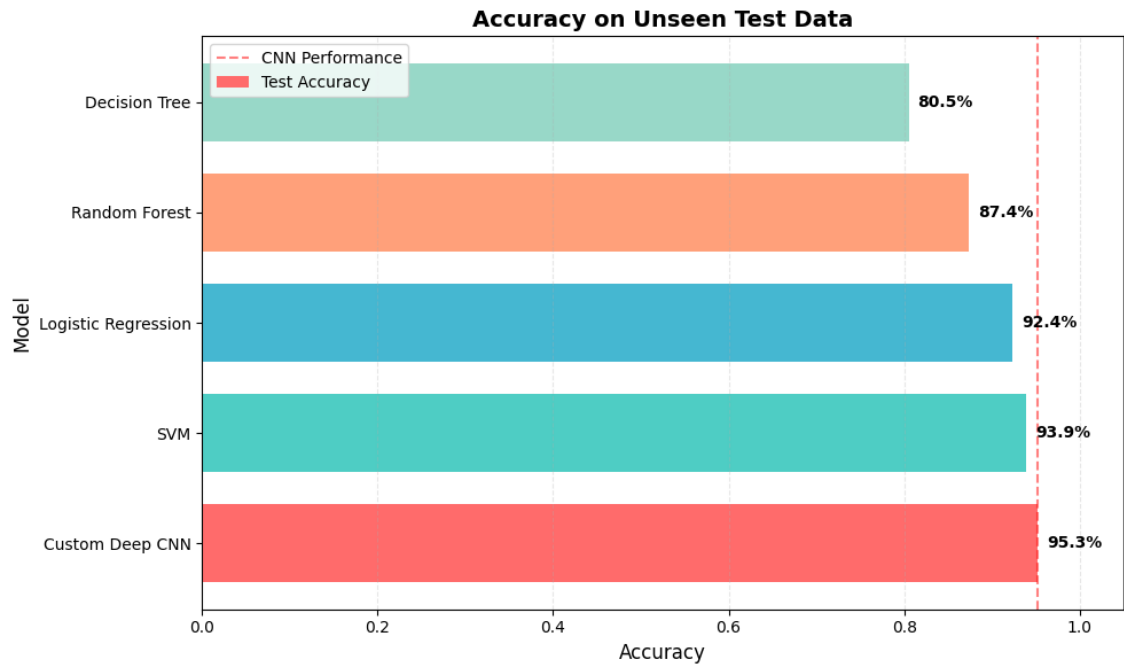


Figure 4.2.1

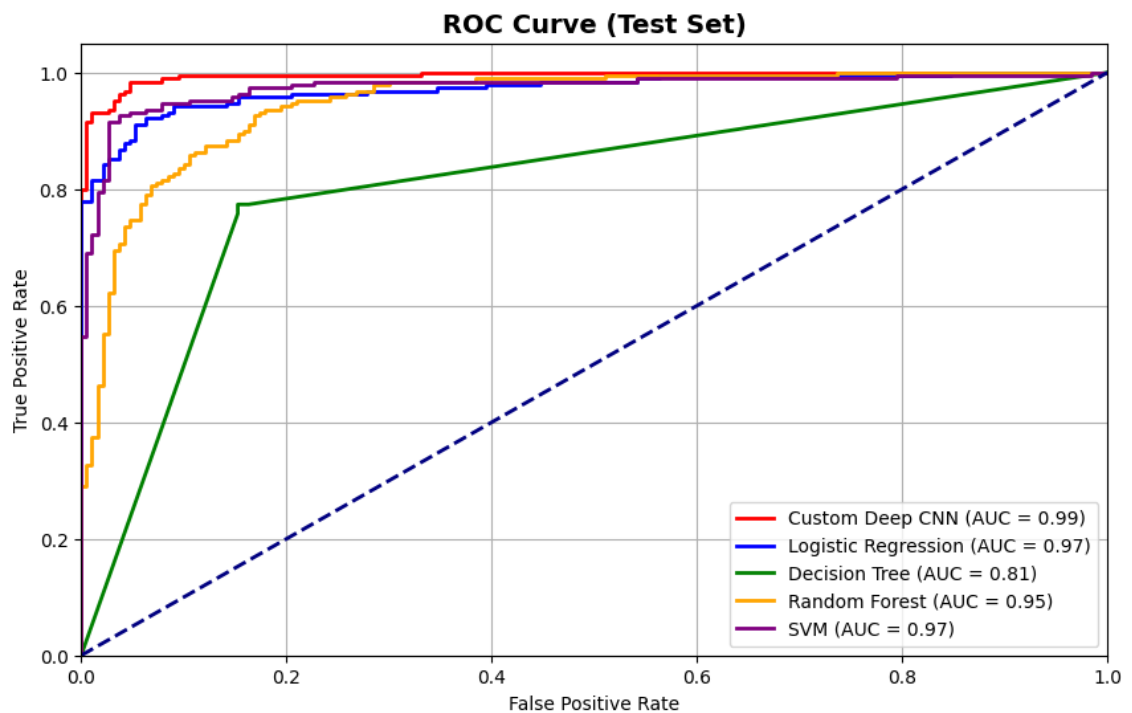


Figure 4.2.2

4.3 Receiver Operating Characteristics

Figure 4.0.3 provides ROC curves for each model. Area-under-curve values reinforce the relative ordering observed in accuracy outcomes. Traditional approaches such as logistic regression and SVM perform reasonably well on the synthetic task, but the CNN shows stronger separation between true-positive and false-positive rates.

The CNN advantage is particularly evident at low false-positive rates, indicating reliable detection when the cost of incorrect ignition alerts is high.

4.4 Interpretation

Taken together, the results reflect three key behaviours:

1. The synthetic patterns used in training are structured, making methods that use spatial feature learning (CNN) more effective.
2. Flattened feature representations constrain traditional models by removing locality information that may be useful for classification.
3. Decision Trees and Random Forests capture useful nonlinear relationships but are limited when the underlying predictors possess spatial coherence that they cannot directly exploit.

4.5 Summary of Findings

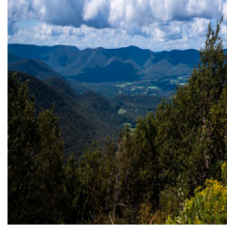
The convolutional architecture produced the most accurate and robust predictions among tested approaches. This improvement is attributed to its ability to learn filters that detect shapes and spatial signatures across the input field, whereas classical models receive pre-flattened vectors without explicit spatial context. These findings highlight the value of convolutional learning in tasks where data contain locality structure, even in simplified or synthetic settings.

File: nofire_0059.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (69.8%)
Logistic NoFire (53.3%)
Decision Fire (100.0%)
Random F Fire (65.7%)
SVM Fire (51.7%)

File: nofire_0918.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (100.0%)
Logistic NoFire (100.0%)
Decision NoFire (100.0%)
Random F NoFire (99.7%)
SVM NoFire (100.0%)

File: fire_0872.jpg
True: Fire



--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (100.0%)
Decision Fire (100.0%)
Random F Fire (83.2%)
SVM Fire (90.3%)

File: nofire_0781.jpg
True: NoFire



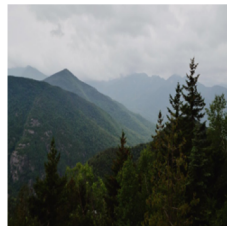
--- PREDICTIONS ---
CNN: NoFire (100.0%)
Logistic NoFire (98.4%)
Decision Fire (100.0%)
Random F NoFire (52.6%)
SVM NoFire (69.3%)

File: fire_0344.jpg
True: Fire



--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (100.0%)
Decision Fire (100.0%)
Random F Fire (97.7%)
SVM Fire (99.9%)

File: nofire_0495.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (100.0%)
Logistic NoFire (100.0%)
Decision NoFire (100.0%)
Random F NoFire (99.6%)
SVM NoFire (100.0%)

File: fire_0217.jpg
True: Fire



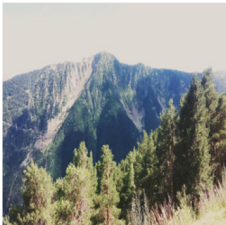
--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (100.0%)
Decision Fire (100.0%)
Random F Fire (95.7%)
SVM Fire (97.5%)

File: fire_0502.jpg
True: Fire



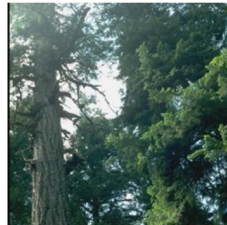
--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (100.0%)
Decision Fire (100.0%)
Random F Fire (75.9%)
SVM Fire (99.0%)

File: nofire_0867.jpg
True: NoFire



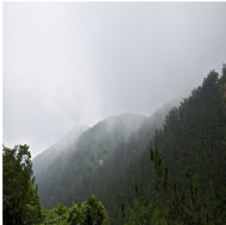
--- PREDICTIONS ---
CNN: NoFire (94.7%)
Logistic NoFire (99.9%)
Decision NoFire (100.0%)
Random F NoFire (83.7%)
SVM NoFire (87.6%)

File: nofire_0039.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (99.9%)
Logistic NoFire (99.9%)
Decision NoFire (100.0%)
Random F NoFire (91.3%)
SVM NoFire (99.1%)

File: nofire_0480.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (99.5%)
Logistic NoFire (98.4%)
Decision NoFire (100.0%)
Random F NoFire (79.5%)
SVM NoFire (96.9%)

File: fire_0935.jpg
True: Fire



--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (99.7%)
Decision Fire (100.0%)
Random F Fire (81.9%)
SVM Fire (99.6%)

File: nofire_0593.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (100.0%)
Logistic NoFire (100.0%)
Decision NoFire (100.0%)
Random F NoFire (91.6%)
SVM NoFire (100.0%)

File: fire_0484.jpg
True: Fire



--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (99.9%)
Decision Fire (100.0%)
Random F Fire (91.3%)
SVM Fire (96.8%)

File: nofire_0499.jpg
True: NoFire



--- PREDICTIONS ---
CNN: NoFire (100.0%)
Logistic NoFire (100.0%)
Decision NoFire (100.0%)
Random F NoFire (97.6%)
SVM NoFire (99.7%)

File: fire_0158.jpg
True: Fire



--- PREDICTIONS ---
CNN: Fire (100.0%)
Logistic Fire (96.7%)
Decision Fire (100.0%)
Random F Fire (79.8%)
SVM Fire (86.7%)

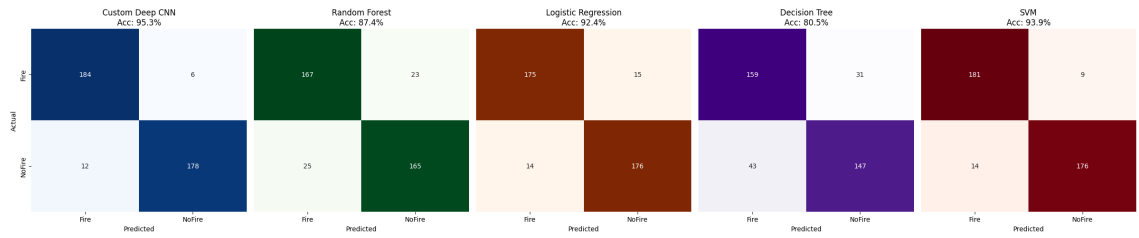


Figure 4.5.2

4.6 Confusion Matrix Analysis

Figure 4.5.2 presents confusion matrices for each evaluated model. These plots provide a more granular view of prediction behaviour beyond overall accuracy.

The convolutional network achieves the strongest separation between classes, with high true-positive and true-negative counts and comparatively few misclassifications. This behaviour aligns with its reported accuracy and demonstrates that the CNN is not only correct more often, but also balanced in how it treats both “Fire” and “NoFire” cases.

Traditional classifiers show mixed trends. The Support Vector Machine performs well, displaying relatively small false-positive and false-negative counts, which is consistent with SVM’s ability to enforce large decision margins. Logistic Regression also yields competitive results but exhibits slightly more confusion between the two classes, reflecting its inherent linear boundary limitations.

Decision Trees produce the weakest confusion structure. Their error pattern shows higher false positives and false negatives, indicating that model splits do not generalise smoothly across the 2-D feature landscape. Random Forest performance improves over Decision Trees, but misclassification levels remain higher than those of the CNN or SVM.

Overall, these matrices reinforce two key interpretations:

1. Although several baseline models achieve strong accuracies, the convolutional network produces fewer ambiguous decisions and a clearer separation between ignition and non-ignition cases.
2. Models relying solely on flattened feature vectors tend to struggle more with boundary uncertainty, while the CNN benefits from learning spatial representations directly from the grid structure.

CONCLUSION

This project evaluated a custom convolutional neural network alongside several classical machine learning algorithms for binary ignition classification on a synthetic spatial dataset. All models achieved relatively strong performance due to the controlled nature of the data; however, clear differences emerged in accuracy, robustness, and decision structure.

The convolutional network consistently outperformed the baselines. This advantage stems from its ability to operate directly on two-dimensional image grids and extract local spatial features through learnable filters. Because ignition patterns in the dataset are spatially organised, the CNN learns relevant neighbourhood structure that traditional models cannot access after flattening. As a result, the CNN exhibited the highest test accuracy and the most balanced confusion matrix, reflecting a strong ability to detect both ignition and non-ignition cases.

Classical models demonstrated mixed performance. Support Vector Machines and Random Forests provided competitive accuracy, as these methods can capture nonlinear boundary relationships. However, their reliance on manually shaped feature vectors limits their ability to leverage spatial context. Decision Trees performed weakest, primarily due to their tendency to fragment decision space, leading to unstable boundaries and higher misclassification. Logistic Regression performed moderately well but was constrained by its linear decision frontier.

Overall, the outcomes highlight two main conclusions:

1. Preserving spatial structure in the input benefits learning performance for pattern recognition problems of this type.
2. Models designed for grid-based feature extraction, such as convolutional networks, are inherently better suited to image-like tasks than classical vector-based classifiers.

Although the synthetic dataset served as a controlled proof-of-concept, the findings suggest that deeper architectures may provide further benefit when applied to more complex or noisy wildfire data. Potential extensions include image augmentation to increase generalisation, multi-class ignition mapping, and integration of additional features such as environmental variables or uncertainty estimates. This work therefore provides a baseline implementation framework and demonstrates the value of spatial learning mechanisms for ignition detection problems.

Bibliography